30	Thorium-232 undergoes the following fission reaction:

$$^{232}_{90}Th \rightarrow {}^{140}_{54}Xe + {}^{90}_{36}Kr + 2\,{}^{1}_{0}n$$

The binding energy per nucleon for the nuclei involved are:

Thorium-232: 7.6 MeV
Xenon-140: 8.3 MeV
Krypton-90: 8.5 MeV

What is the energy released by this fission reaction?

A	1.64×10^{8} J	**B**	2.63×10^{-11} J	**C**	1.47×10^{-12} J	**D**	2.63×10^{-17} J

Answer: B

Explanation/Working:

Thorium-232 has 232 nucleons.

Binding energy = 232 × 7.6 = 1763.2 MeV

Xenon-140: 140 × 8.3 = 1162.0 MeV

Krypton-90: 90 × 8.5 = 765.0 MeV

	Energy released = Binding energy of products − Binding energy of reactants = 1927.0 − 1763.2 = 163.8 MeV 1 MeV = 1.602×10^{-13} J Energy in joules = $163.8 \times 1.602 \times 10^{-13} = 2.63 \times 10^{-11}$ J

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